

**IS (102) Data Modeling in Software Engineering
Project Report**

Online Novel Reading and Writing System



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Chapter 1

Introduction

1.1 Abstract

In today's digital age, the consumption of literature has shifted from physical books to dynamic, accessible web platforms. The "Online Novel Reading and Writing System" is a comprehensive web-based application designed to bridge the gap between aspiring authors and avid readers. This system provides a centralized hub where users can explore diverse genres, publish multi-chapter stories, and engage in a social literary community. By leveraging modern web technologies like PHP and MySQL, the platform ensures role-based access control and secure data management, creating a reliable environment for creative expression and digital reading.

1.2 Project Aim

The aim of this project is to develop a robust, user-friendly digital library and publishing tool that encourages literacy and creativity. Specifically, the project seeks to:

- Provide a seamless interface for readers to discover and navigate novels with customizable reading settings.
- Empower writers with a dashboard to perform CRUD operations on their intellectual property.
- Foster community interaction through "Follow" mechanisms and star-based rating systems.
- Demonstrate core Software Engineering principles through organized system requirements and modular design.

Chapter 2

Statement of Problem

2.1 Business Needs

The main purpose of this request is to build an **Online Novel Reading and Writing System** that addresses the needs of digital readers and writers. These needs include:

- 1) Providing a centralized platform for browsing and reading novels online.
- 2) Allowing users to create, publish, and manage their own multi-chapter stories.
- 3) Enabling social interactions such as following authors and rating novels.
- 4) Ensuring secure authentication, user privacy, and administrative moderation.

2.2 Business Requirements

Regarding the problems mentioned previously, we want a platform that possesses the following capabilities:

- 1) User registration, login, and profile management.
- 2) Browsing and searching novels by title, genre, and status (Ongoing/Completed).
- 3) Reading novels with chapter navigation and adjustable font size.
- 4) Bookmarking novels for quick access later.
- 5) Writing and managing stories and chapters (CRUD operations) by the same user account.
- 6) User interaction features including following authors and rating novels.
- 7) Administrative access to moderate content, manage users, and monitor system statistics.

2.3 Business Values

By building this system, we, the developers, can achieve a platform that increases user retention through personalized content and establishes a reputable digital brand for literature. For the users, it provides high intangible value by lowering the barrier to entry for new authors and providing free, high-quality entertainment for readers.

2.4 Expected Solution

Above all business needs and requirements, the following constraints must be placed while the system is being constructed:

- 1) **Platform Accessibility** – The system must be web-based and compatible with modern browsers and devices.
- 2) **Data Security** – User data, stories, and ratings must be securely stored and protected.
- 3) **Content Management** – Authors can only edit/delete their own content; administrators moderate the platform.
- 4) **Usability** – The interface must be intuitive, simple, and consistent for all users.
- 5) **Performance** – Pages should load quickly, and searches and chapter navigation must be efficient.

Chapter 3

Scenario-based Modeling

The Online Novel Reading and Writing System leverages scenario-based modeling to define and analyze different user interactions, ensuring an efficient and user-friendly platform for literary creation and consumption. This model includes various scenarios such as readers discovering and bookmarking novels, writers publishing new chapters, administrators moderating content, and users interacting through follows and ratings. Use case diagrams are used to visualize system behavior under different conditions, improving design accuracy and system validation. By considering multiple real-world scenarios, including normal reading, content editing, and administrative interventions, this approach enhances the system's reliability, usability, and performance. The scenario-based modeling technique ensures a structured and intuitive design, making novel reading and writing seamless for readers, writers, and administrators.

3.1 Use cases

Actor: Admin

User Profiles:

- Authenticate using role-based access control.
- View and manage user accounts.
- Content moderation (approve or remove stories and comments).
- Monitor system usage and platform statistics.
- Ensure compliance with platform rules and content policies.

Actor: Reader

User Profiles:

- Register, log in, and manage personal profile
- Browse and search novels by title, author, or genre.
- Read novels with chapter navigation and font customization.
- Bookmark novels for personal library management.
- Interact with novels through ratings and follow other users.

Actor: Writer (Author)

User Profiles:

- Register and authenticate accounts.
- Manage story lifecycle: Create, read, update, and delete (CRUD) stories and chapters.
- Personal dashboard for managing published content.
- View reader interactions and ratings on personal stories.
- Update author profile information.

Use Case - 1: User Registration and Authentication

Scenario:

1. User Registration

- The user navigates to the registration page.
- The user provides a username, email, and password.
- The system validates the credentials against the MySQL database.

2. User Login

- The user enters their username and password on the login form.
- The system establishes a secure user session upon successful validation.

Use Case - 2: Novel Browsing and Search

Scenario:

1. Browsing Novels

- The user navigates to the central browsing page to view all available novels.
- The user can filter novels by genre or story status, such as Ongoing or Completed.

2. Searching Novels

- The user enters specific title keywords into the search bar.
- The system displays a filtered list of novels retrieved from the database.

Use Case - 3: Reading System & Customization

Scenario:

1. Chapter Navigation

- The reader selects a specific chapter to read.
- The reader uses "Next" and "Previous" buttons to navigate between chapters.

2. Text Customization

- The reader adjusts the font size to their preference.
- The system applies the selected font settings to the chapter display.

Use Case - 4: Bookmark Management

Scenario:

1. Adding Bookmarks

- The reader navigates to a specific novel's detail page.
- The reader clicks the "Bookmark" button.

2. Accessing Library

- The system adds the novel to the user's dedicated bookmark section for quick access.

Use Case - 5: Writing and Story Management (CRUD)

Scenario:

1. Creating a Story

- The author inputs story metadata including title, description, and genre via the **Create Story** function.
- The author writes and saves the initial text content for chapters using the **Add Chapter** use case.
- The author can choose to **Draft Story** or **Draft Chapter** to save progress privately without public visibility.

2. Managing Content

- The author performs Update (**Edit Story/Chapter**) or **Delete** operations on their existing content.
- The author can **Publish Story** or **Publish Chapter** to make the content accessible to the general user base.
- The author can **Change Status** (e.g., from "Ongoing" to "Completed") to update the story's lifecycle.
- The system restricts these permissions strictly to the original author, though the **Admin** retains the ability to **Delete Story** for moderation.

Use Case - 6: User Follow System

Scenario:

1. Following a User

- The user visits another member's public profile page.
- The user clicks the "Follow" button.

2. Updating Social Stats

- The system records the follow relationship in the database.
- The system automatically updates the follower and following counts for both users.

Use Case - 7: Novel Rating System

Scenario:

1. Providing a Rating

- An authenticated user selects a star rating on a 1-5 scale on a novel's detail page.

2. Calculating Averages

- The system stores the individual rating in the database.
- The system updates and displays the overall average rating for that novel.

Use Case - 8: User Search and Public Profiles

Scenario:

1. Finding Members

- The user enters a target @username into the system's search bar.

2. Viewing Profiles

- The system displays the selected user's public profile information.
- The system lists all published works authored by that user.

Use Case - 9: Administrative Moderation

Scenario:

1. User Management

- The administrator logs in with privileged credentials.
- The admin selects specific user accounts to suspend or ban for rule violations.

2. Content Moderation

- The admin reviews reported or inappropriate stories.
- The admin deletes content that does not comply with platform policies.

3.2 Use Case Diagram

Here is the use case diagram for **Online Novel Reading and Writing System**:

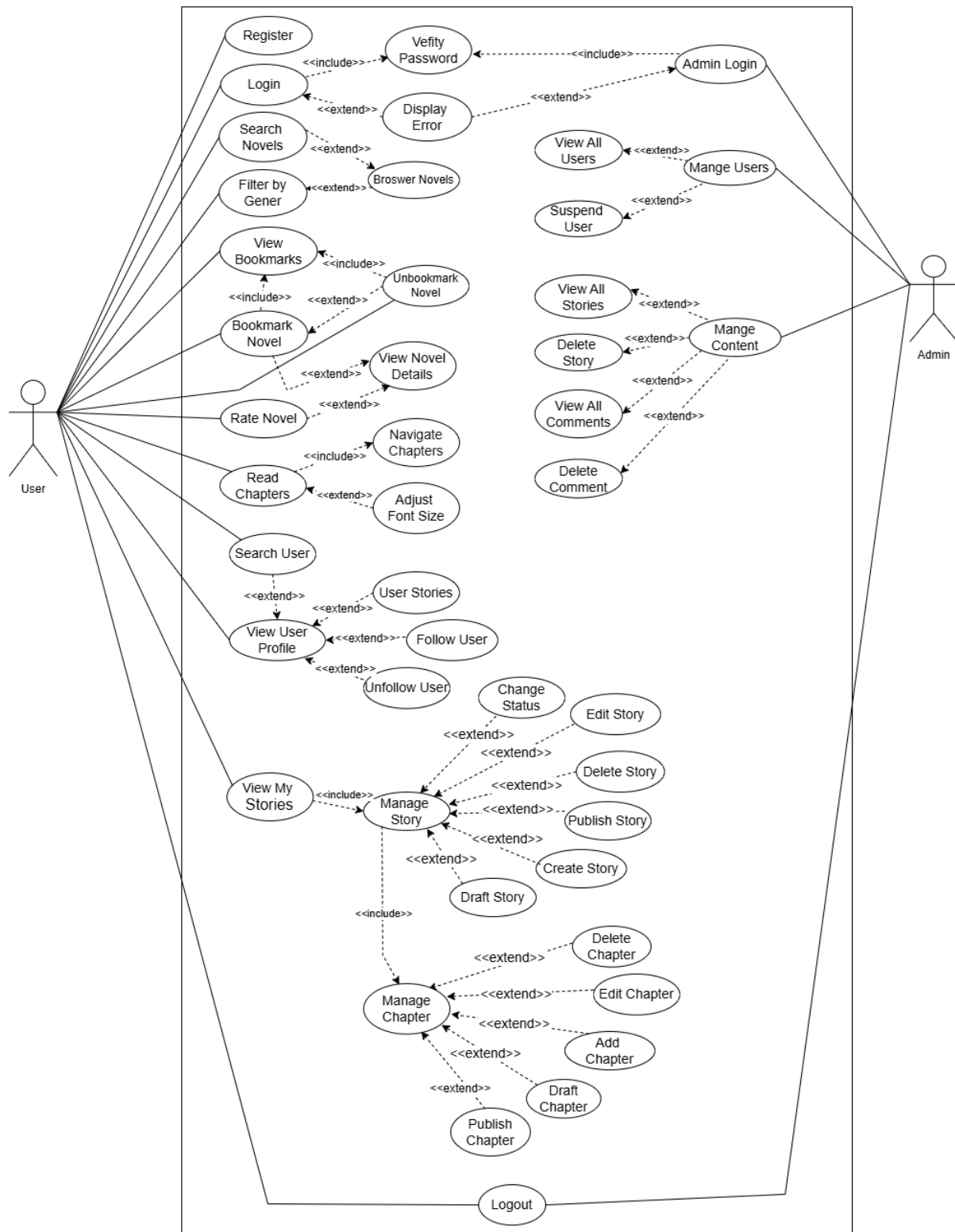


Figure 3.1: Use case diagram for Online Novel Reading and Writing System

Chapter 4

Flow Oriented Modeling

A Data Flow Diagram (DFD) is a graphical representation that depicts the flow of data through the Online Novel Reading / Writing System. It visually shows how data moves between different components, such as users, administrators, and the system's internal processes, helping understand how the system functions.

The Online Novel Reading / Writing System DFD is created at different levels of detail to provide a clear overview of data management:

4.1 Context Diagram

The Context Diagram focuses on the high-level interactions between the Online Novel Reading / Writing System and external entities. It defines the entire system as a single process (Process 0) and identifies the data flows to and from external entities such as USER (READER / WRITER) and ADMIN.

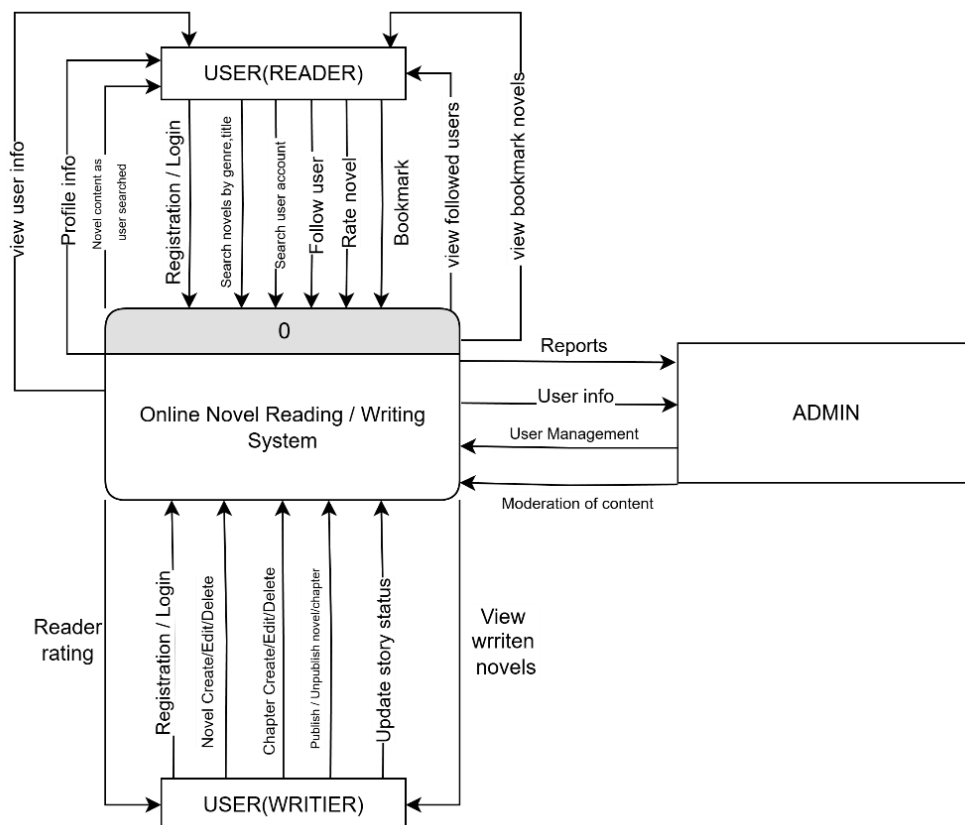


Figure 4.1: Context diagram for Online Novel Reading / Writing Model

4.2 Level 0 Diagram

The Level 0 Diagram provides a more detailed breakdown of the system by decomposing the single process from the Context Diagram into sub-processes. It shows how internal data stores, such as Users Data (D1), Novel Data (D2), and Interaction Data (D3), interact with the main functional processes.

Here's a breakdown of the key elements and processes in the Level 0 diagram:

- 1.0 Manage User Accounts
- 2.0 Search & Browse Novels
- 3.0 Reading System
- 4.0 Manage Stories & Chapters
- 5.0 Manage Bookmarks & Ratings
- 6.0 Platform Moderation

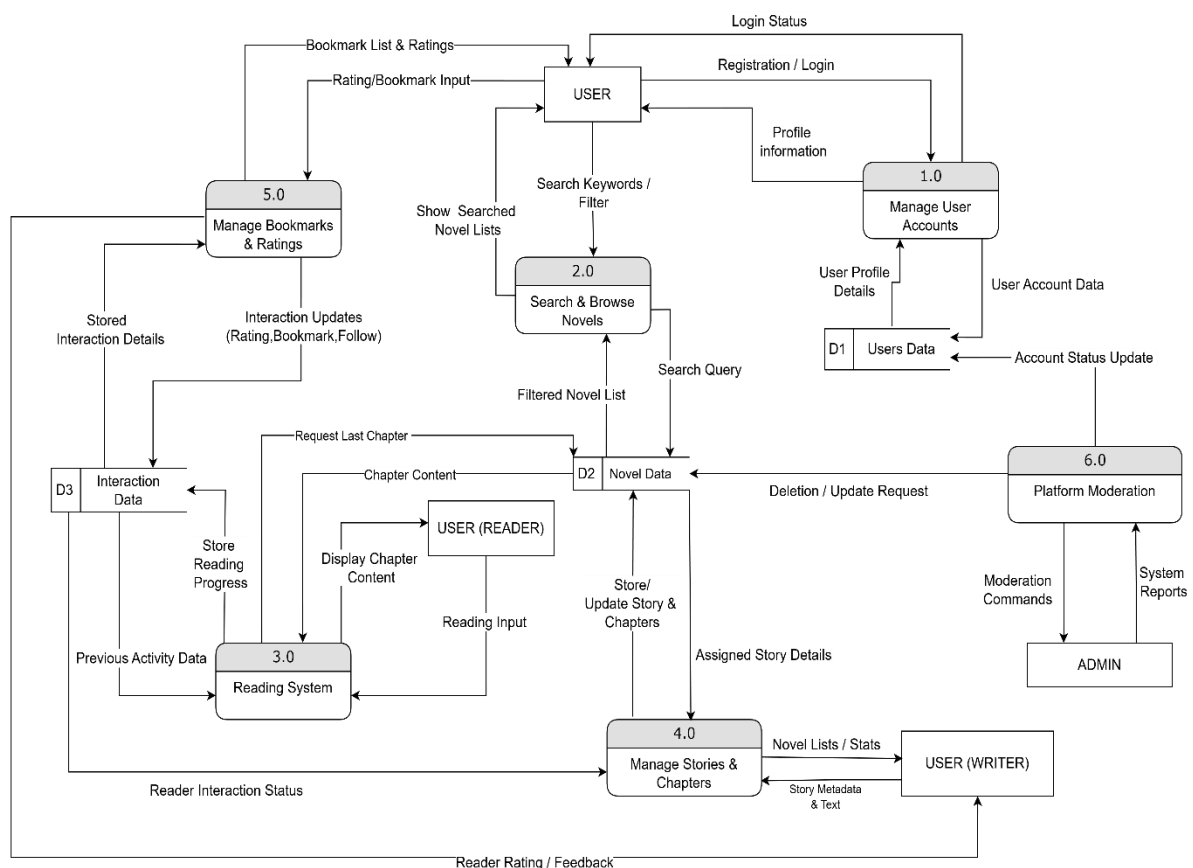


Figure 4.2: Level 0 diagram for Online Novel Reading / Writing System

4.3 Level 1 Diagram

Level 1 diagrams break down complex Level 0 processes to show detailed data handling logic.

4.3.1 Process 1.0: Manage User Accounts

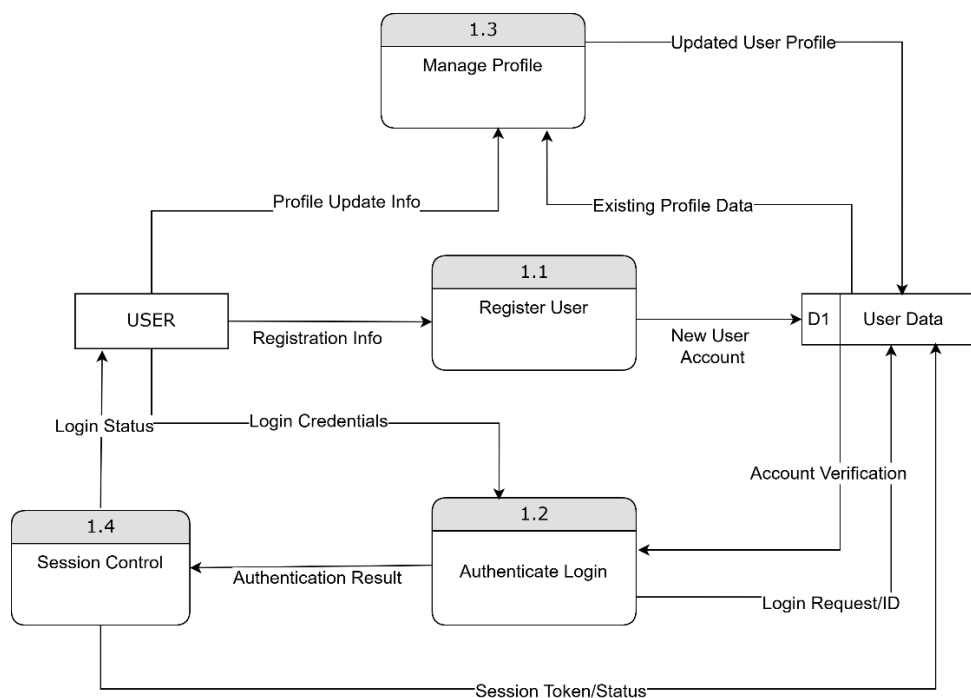
This process handles user security , registration and profile personalization.

1.1 Register User: Creates new accounts in D1.

1.2 Authenticate Login: Validates credentials against D1.

1.3 Manage Profile: Updates user profile info in D1.

1.4 Session Control: Delivers Login Status to the User



**Figure 4.3: Figure 4.3.1: Process 1.0 Diagram of
Level 1 diagram for Online Novel Reading / Writing System**

4.3.3 Process 3.0: Reading System

3.1 Fetch Content: Retrieves chapters from D2 for display.

3.2 Track Progress: Monitors Reading Input to calculate progress.

3.3 Record Logs: Saves reading history into D3.

3.4 Resume State: Recalls the Resume Point from D3.

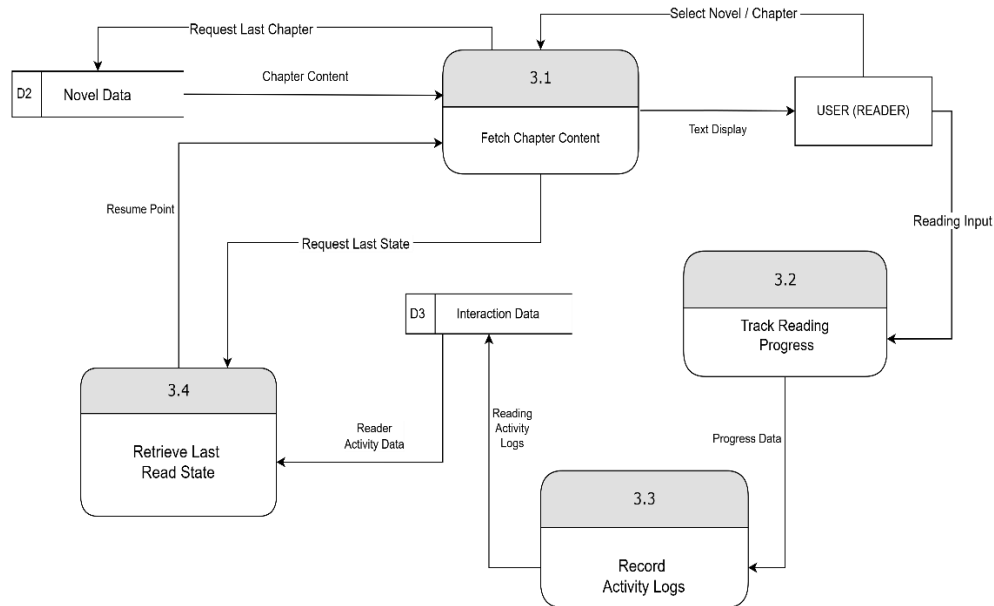


Figure 4.3.3: Process 3.0 Diagram of Level 1 diagram for Online Novel Reading / Writing System

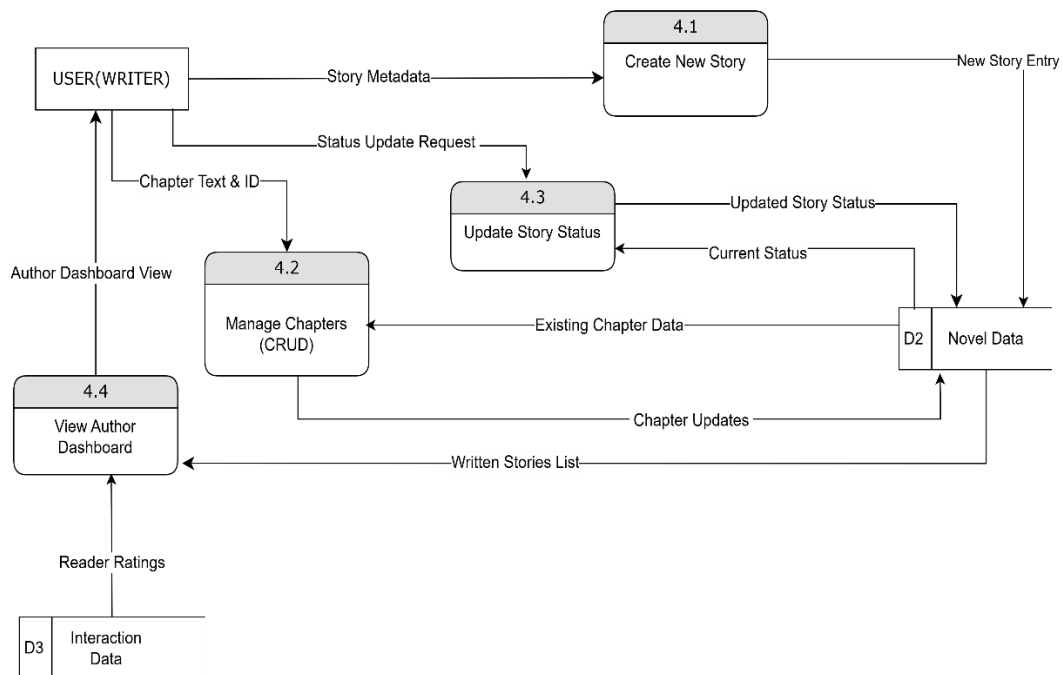
4.3.4 Process 4.0: Manage Stories & Chapters

4.1 Create Story: Saves metadata to D2 as a new entry.

4.2 Manage Chapters: Updates D2 with new chapter text.

4.3 Update Status: Changes story availability in D2.

4.4 View Dashboard: Shows story stats and D3 ratings to the Writer.



**Figure 4.3.4: Process 4.0 Diagram of
Level 1 diagram for Online Novel Reading / Writing System**

Chapter 5

Conceptual Data Modeling

The foundation of the Online Novel Reading and Writing Platform's data storage lies in conceptual data modeling, specifically through the Entity-Relationship (ER) diagram. This diagram provides a visual structure of the system by identifying the main entities and the relationships between them.

The primary entities in this system include User, Novel, Chapter, Genre, Follow, Rating, and Bookmark. The ER diagram shows how a user can create novels, a novel can contain multiple chapters, and users can follow other users, rate novels, and bookmark stories.

By defining these entities and their relationships, the ER diagram helps organize data efficiently, maintain consistency, and ensure accurate retrieval of information within the platform.

Here is the ER diagram of the Online Novel Reading and Writing Platform.

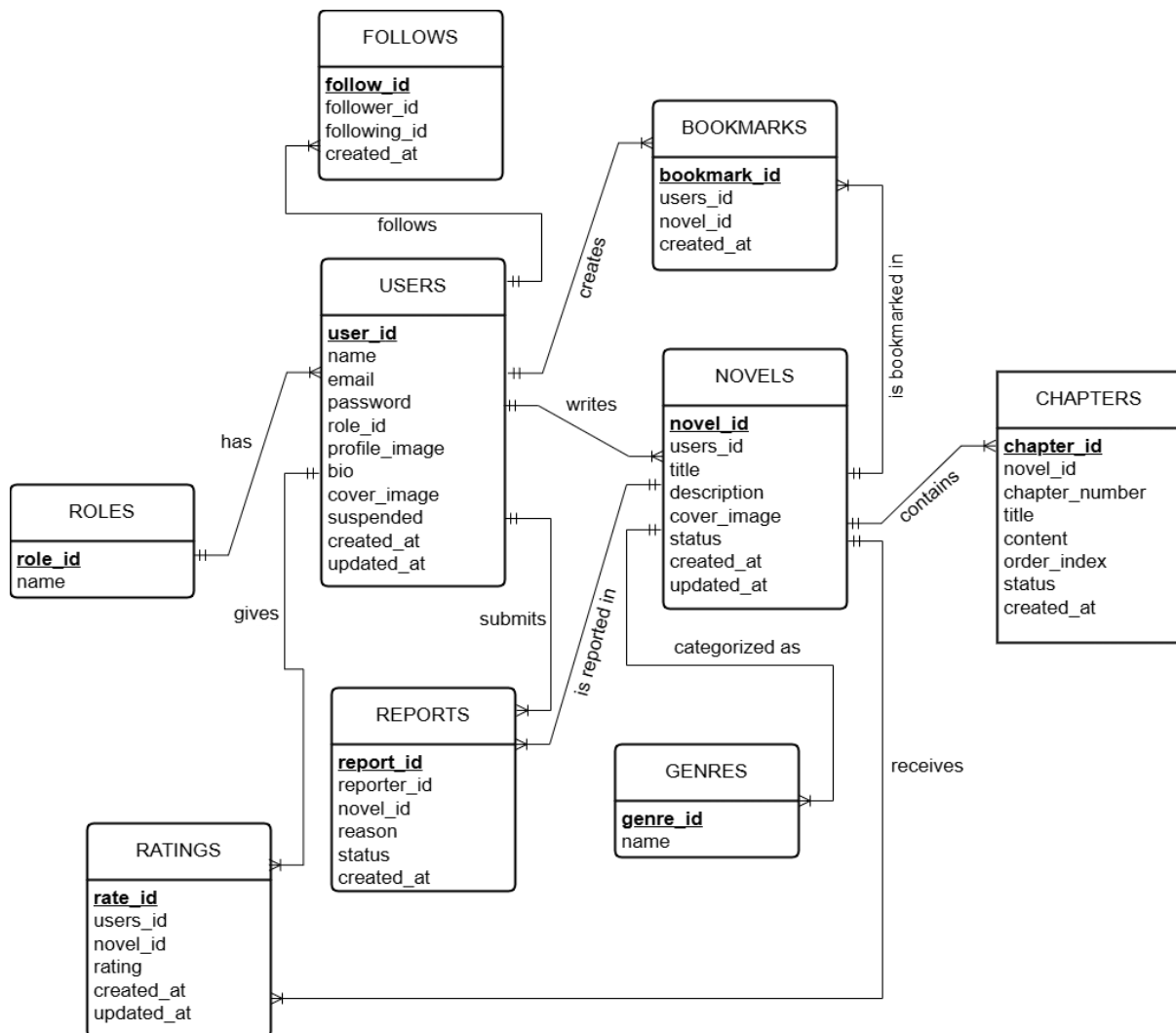


Figure 5.1: ER Diagram of Online Novel Reading and Writing System

Chapter 6

Class Oriented Modeling

At the core of the Online Novel Reading and Writing Platform is a structured interaction between system components. To represent this structure, a Class Diagram is used in class-oriented modeling.

The class diagram, created using UML, illustrates the main classes of the system along with their attributes and methods. The key classes include User, Novel, Chapter, Genre, Follow, Rating, and Bookmark.

The diagram shows how these classes are related. For example, a User can create multiple Novels, a Novel can contain multiple Chapters, and a User can follow other users and rate or bookmark novels.

This modeling approach provides a clear blueprint of the system structure and supports better understanding, maintenance, and future enhancement of the platform.

Here is the Class Diagram of the Online Novel Reading and Writing Platform.

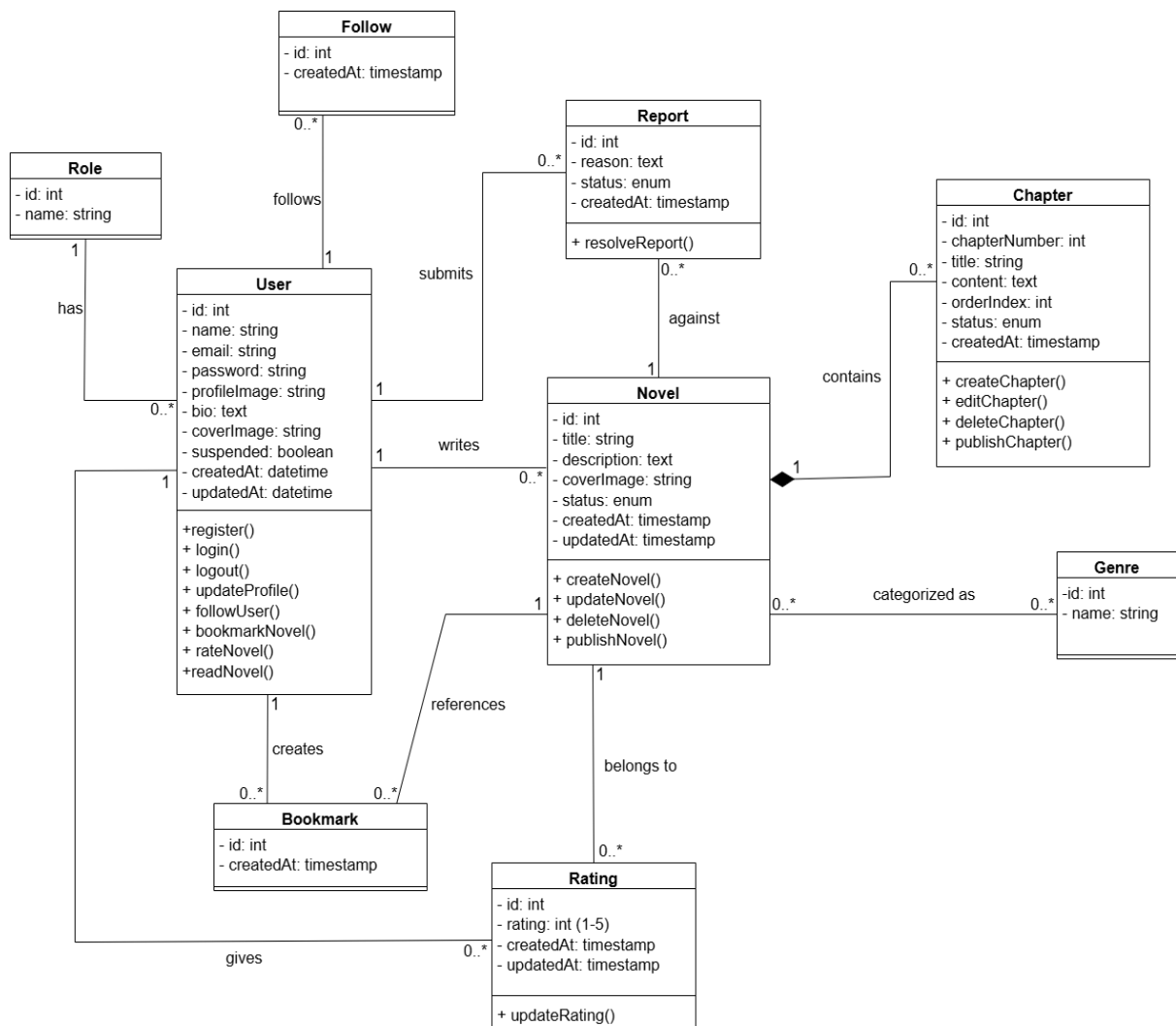


Figure 6.1: Class Diagram of Online Novel Reading and Writing Platform

Chapter 7

Behavioral Modeling

The Online Novel Reading and Writing System employs sequence diagrams to illustrate dynamic interactions between actors and system components over time. These diagrams complement the use case scenarios by showing the specific message sequences to accomplish system functions.

Sequence diagram For User Authentication

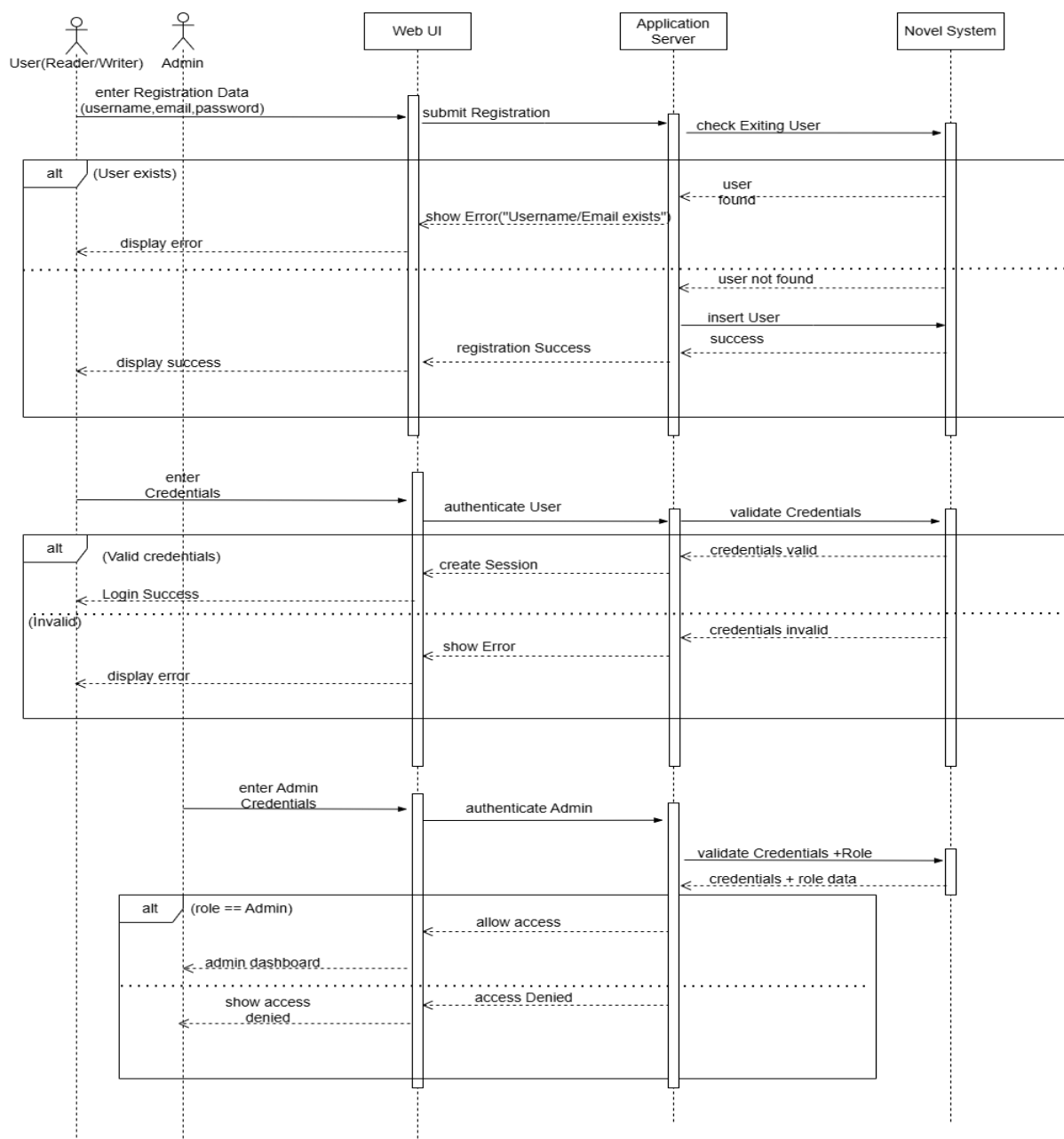


Figure 7.1 : Sequence diagram for User Authentication

Sequence diagram For Reading Operations

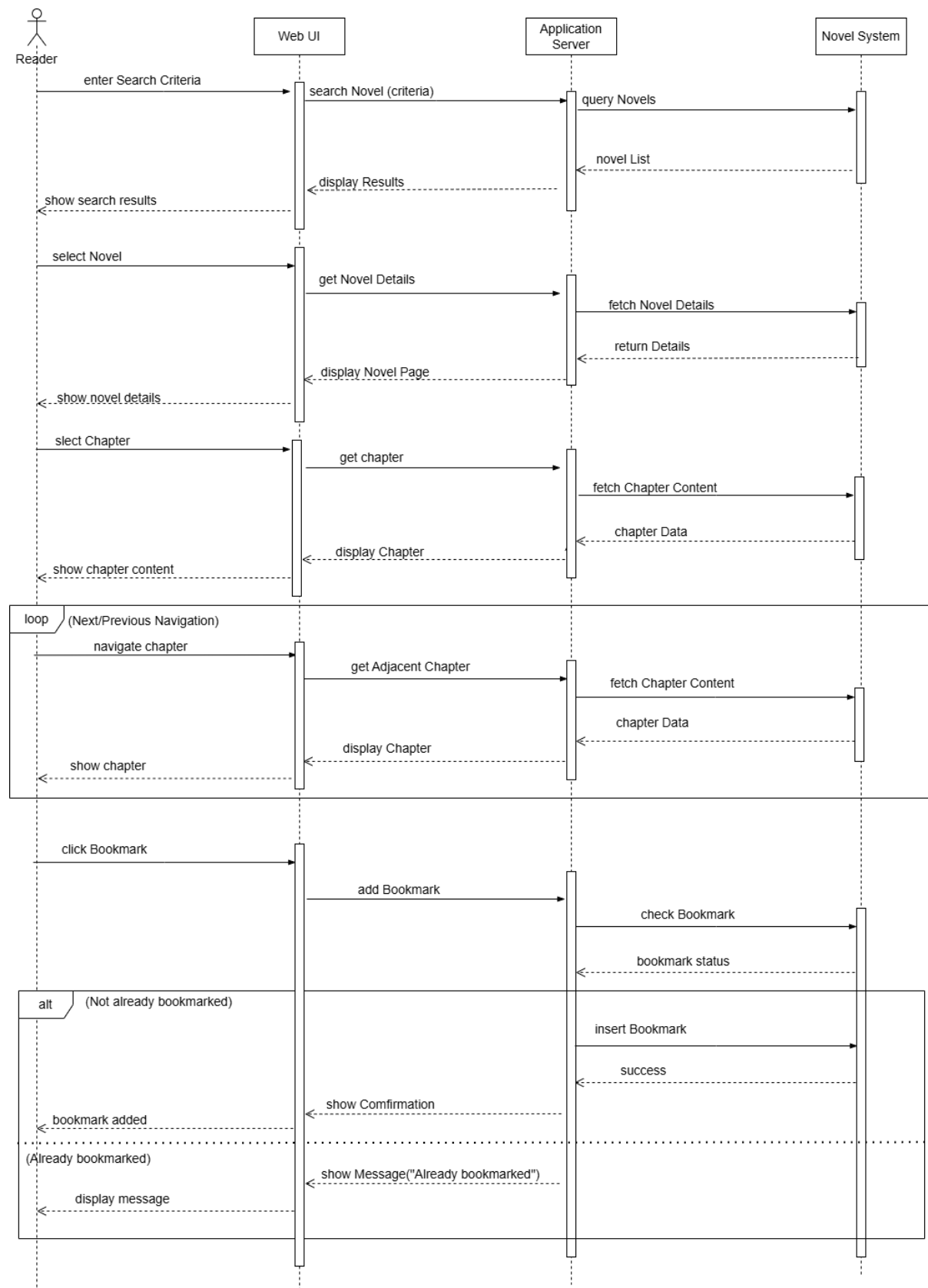


Figure 7.2 : Sequence diagram for Reading Operations

Sequence diagram For Writing Operations

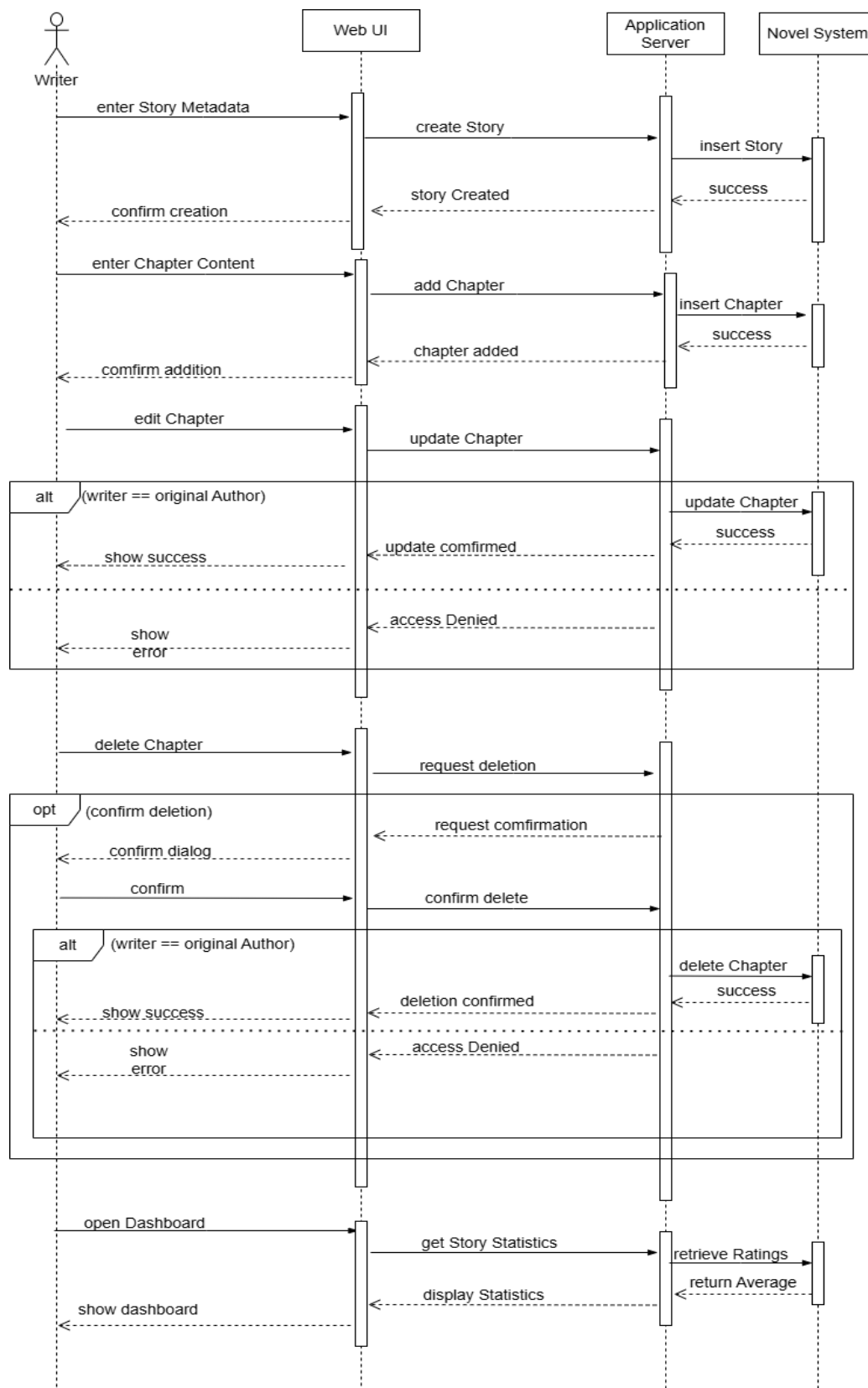


Figure 7.3 : Sequence diagram for Writing Operations

Sequence diagram For User Interaction

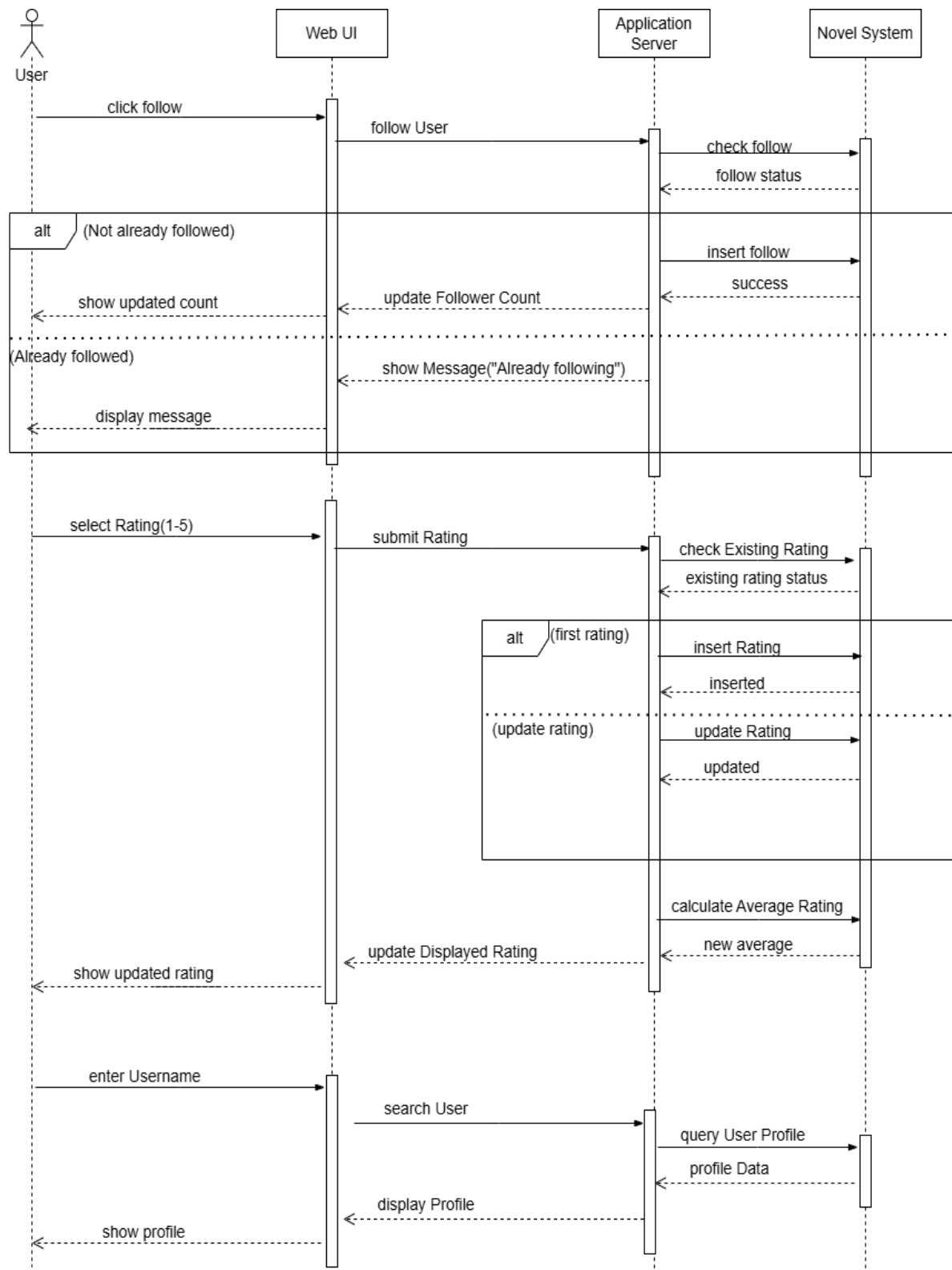


Figure 7.4 : Sequence diagram for User Interaction

Sequence diagram For Admin Operations

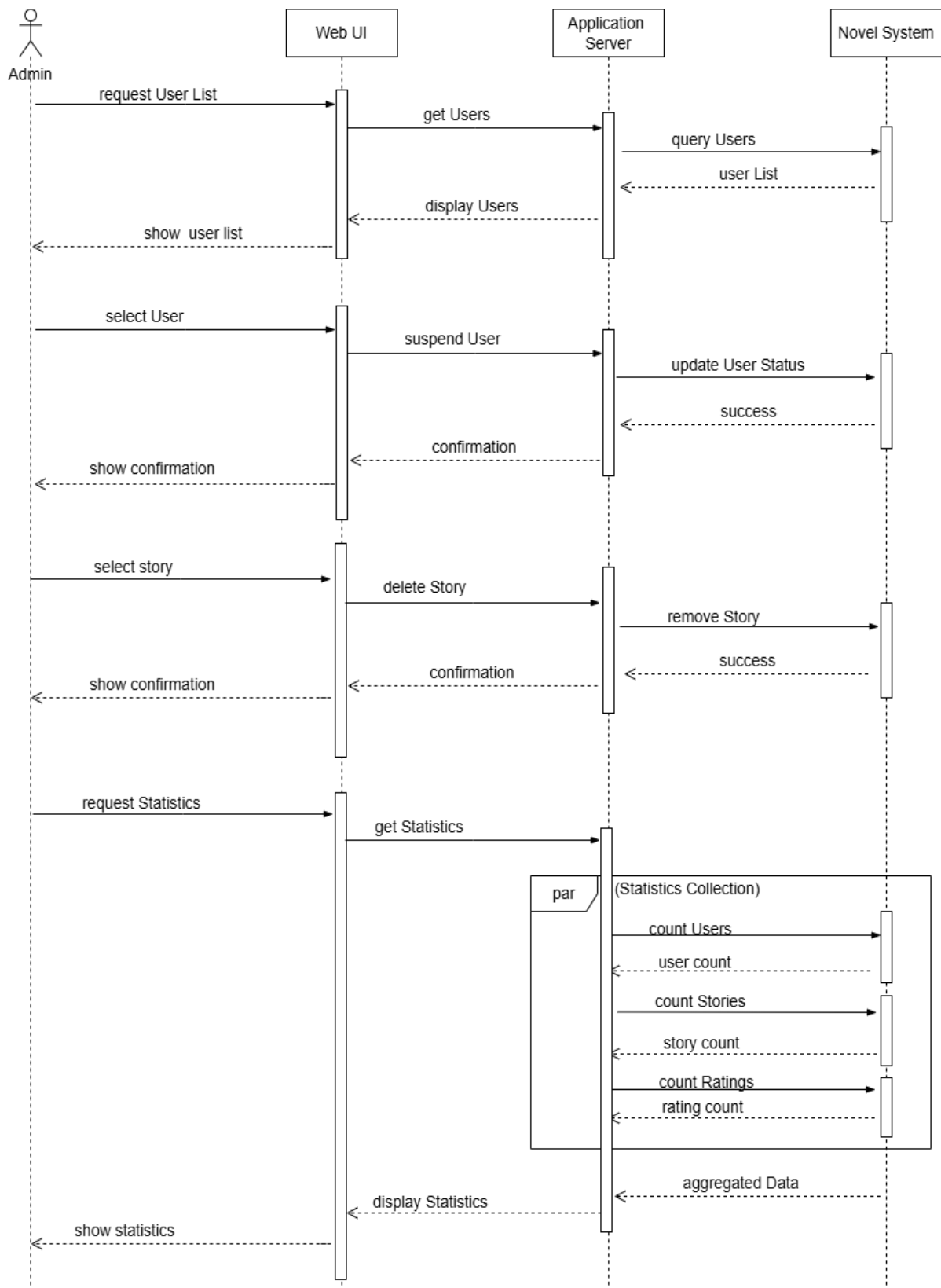


Figure 7.5 : Sequence diagram for Admin Operations

Implementations

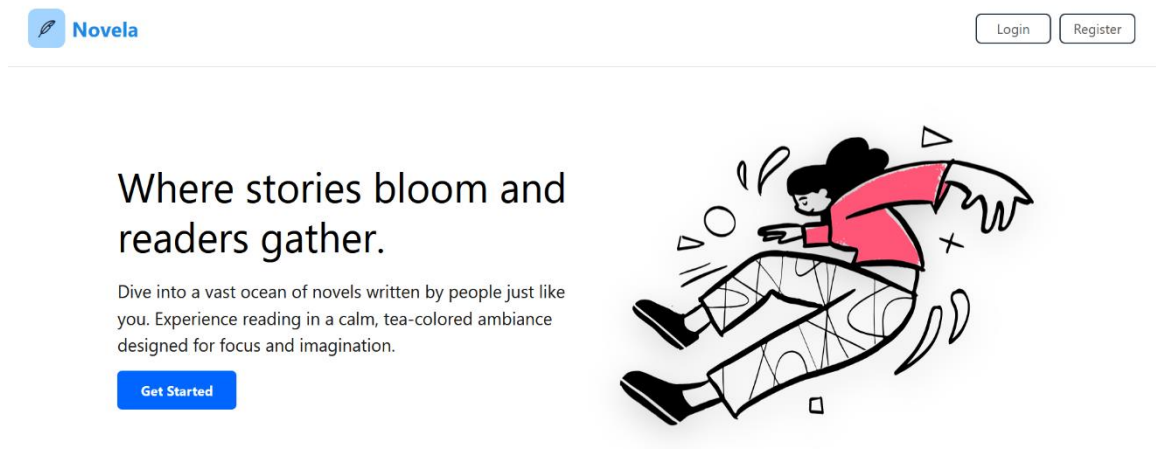


Figure 1: Welcome Page

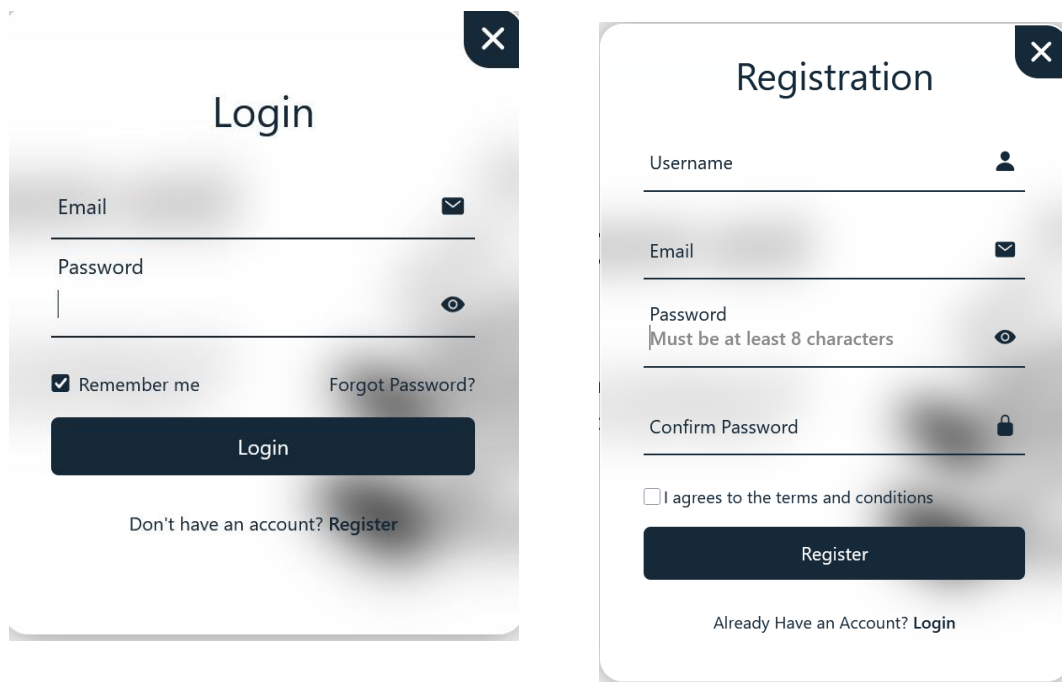
The image displays two mobile app screens side-by-side. The left screen is titled "Login" and features a close button (X) in the top right corner. It has two input fields: "Email" with an envelope icon and "Password" with an eye icon. Below these fields are a checked "Remember me" checkbox and a "Forgot Password?" link. A large dark blue "Login" button is at the bottom, with a link "Don't have an account? Register" below it. The right screen is titled "Registration" and also has a close button (X) in the top right corner. It features three input fields: "Username" with a person icon, "Email" with an envelope icon, and "Password" with an eye icon. The password field has a hint "Must be at least 8 characters". Below the password field is a "Confirm Password" field with a lock icon. At the bottom of the form is an unchecked checkbox "I agree to the terms and conditions" and a large dark blue "Register" button. Below the button is a link "Already Have an Account? Login".

Figure 2: Login Form and Register Form

All Published Stories

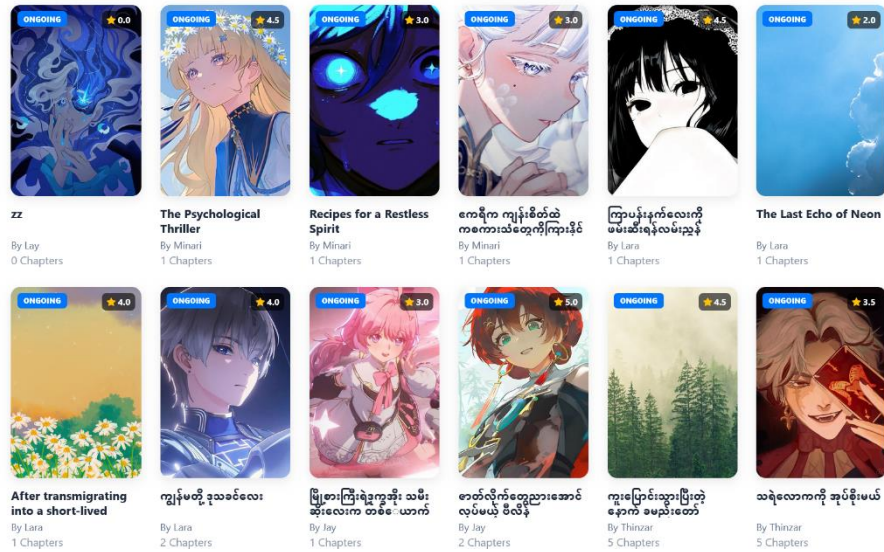


Figure 3: Home Page

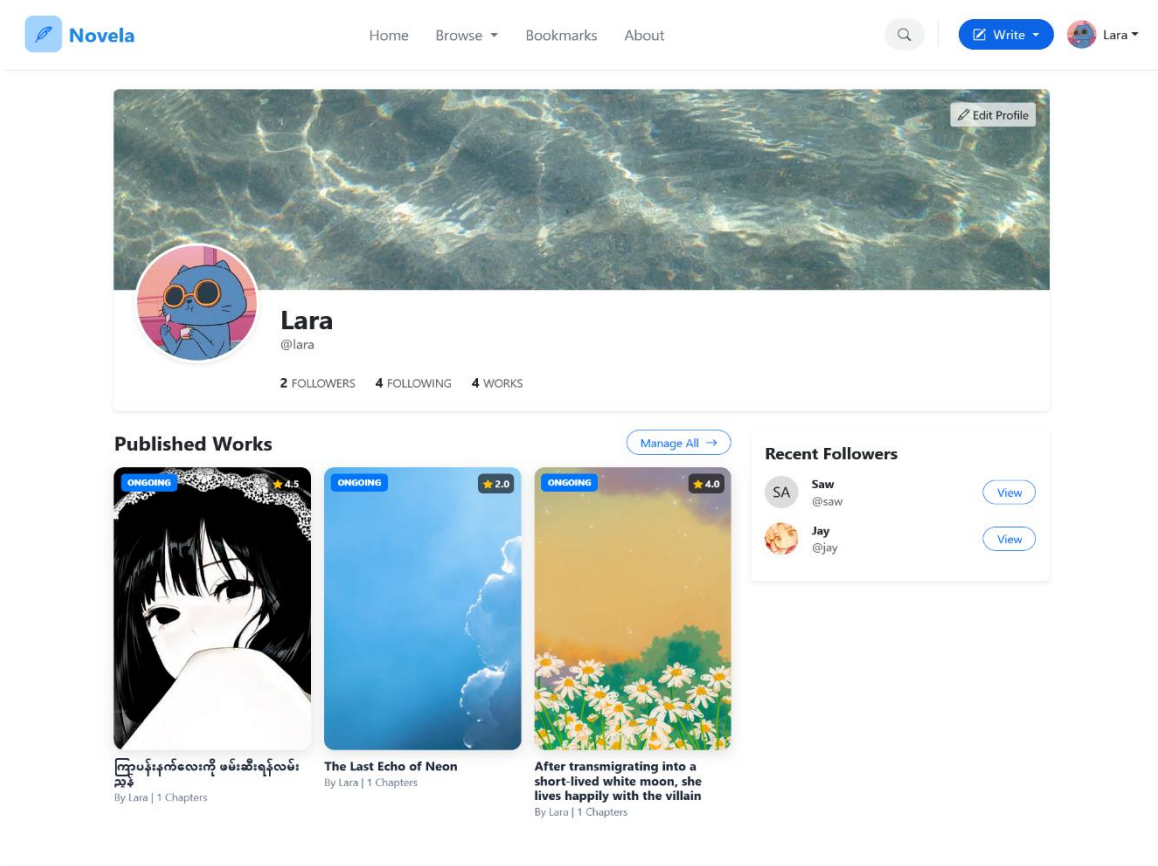


Figure 4: Profile Page

< Back

Edit Profile

About You (Bio)

Tell your readers about yourself...

Profile Picture

Browse... No file selected.

Recommended: Square image (500x500px)

Cover Image

Browse... No file selected.

Recommended: Wide image (1200x300px)

Cancel [Save Changes](#)

Figure 5: Edit Profile

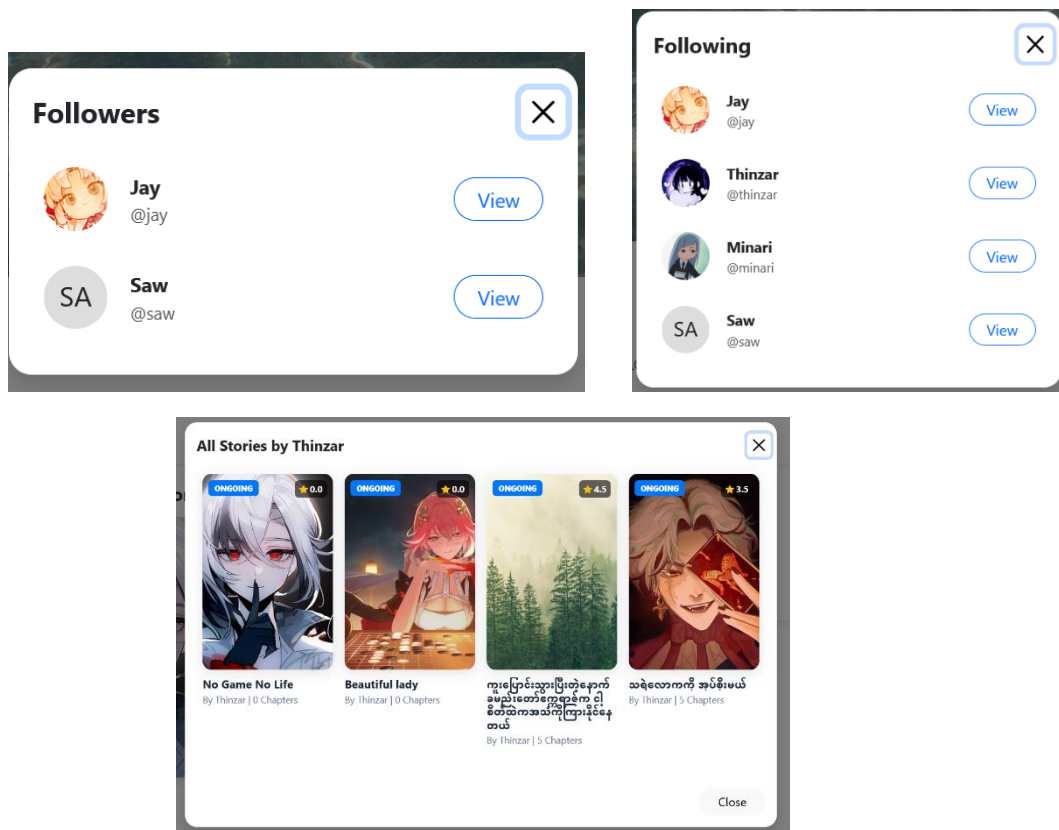


Figure 6: View Follower, Following and published stories of other users pop-ups

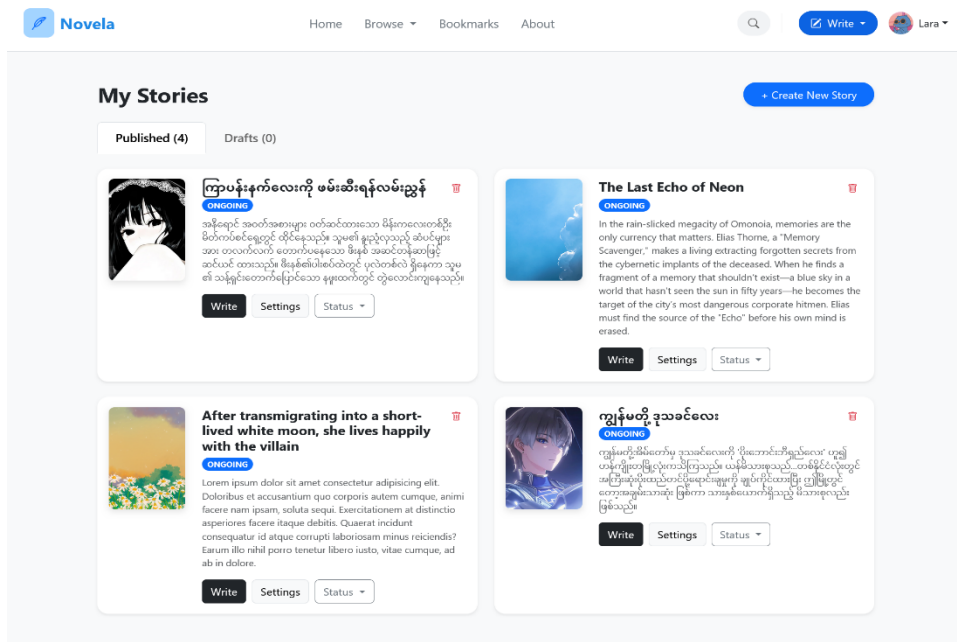


Figure 7: My stories page (For Writers)

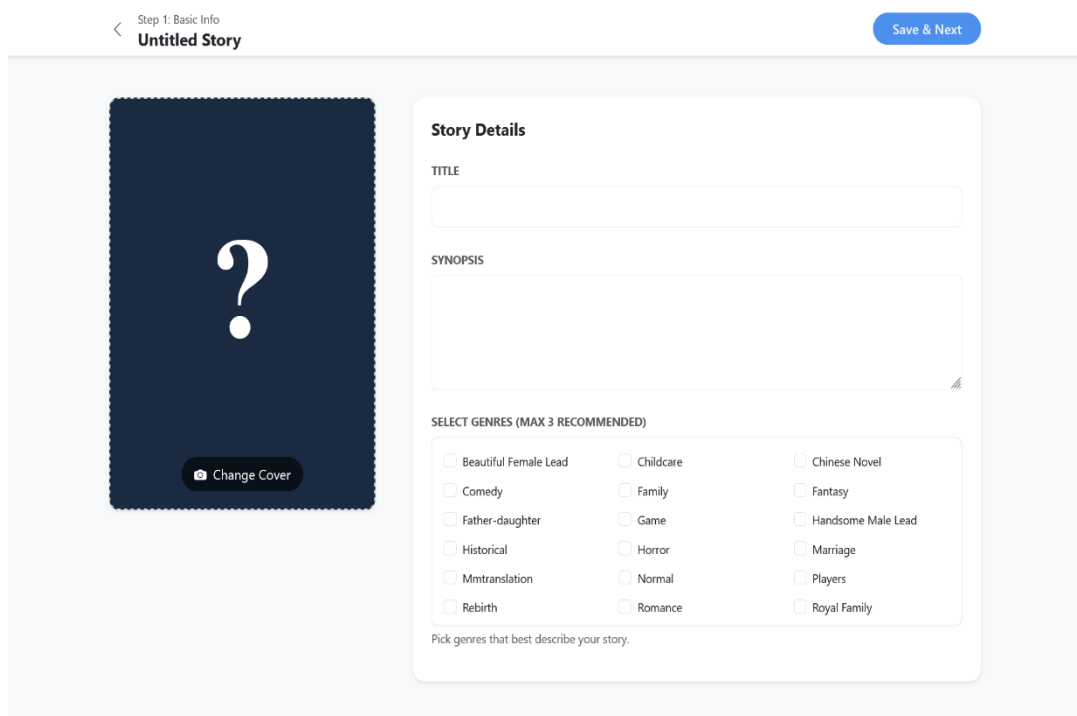


Figure 8: Create New Stories (For Writers)

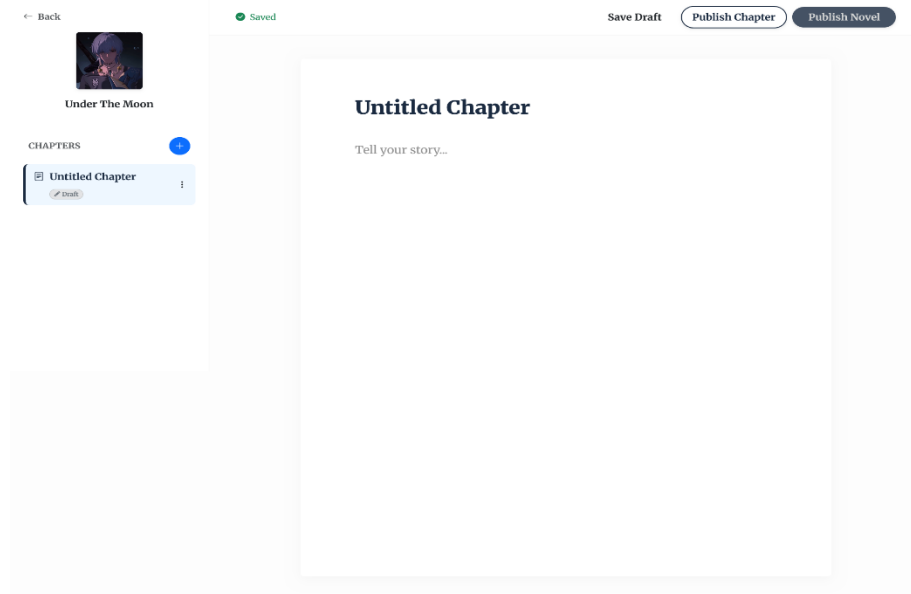


Figure 9: Create New Stories-Continue (For Writers)

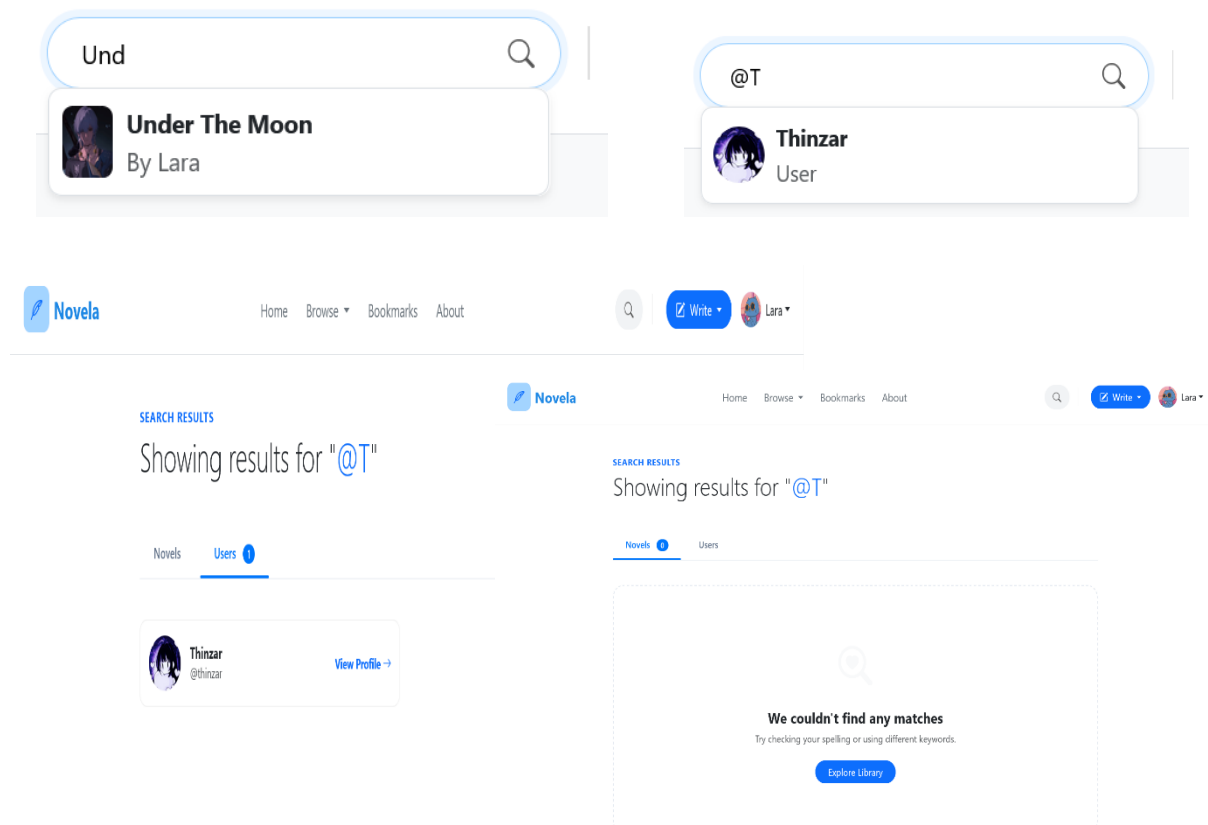


Figure 10: Searching

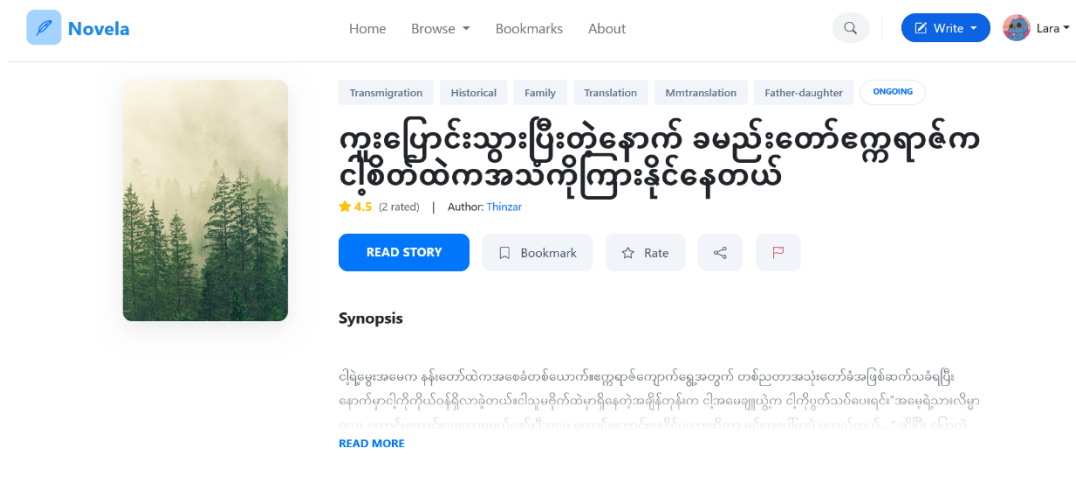


Figure 11: View Novel

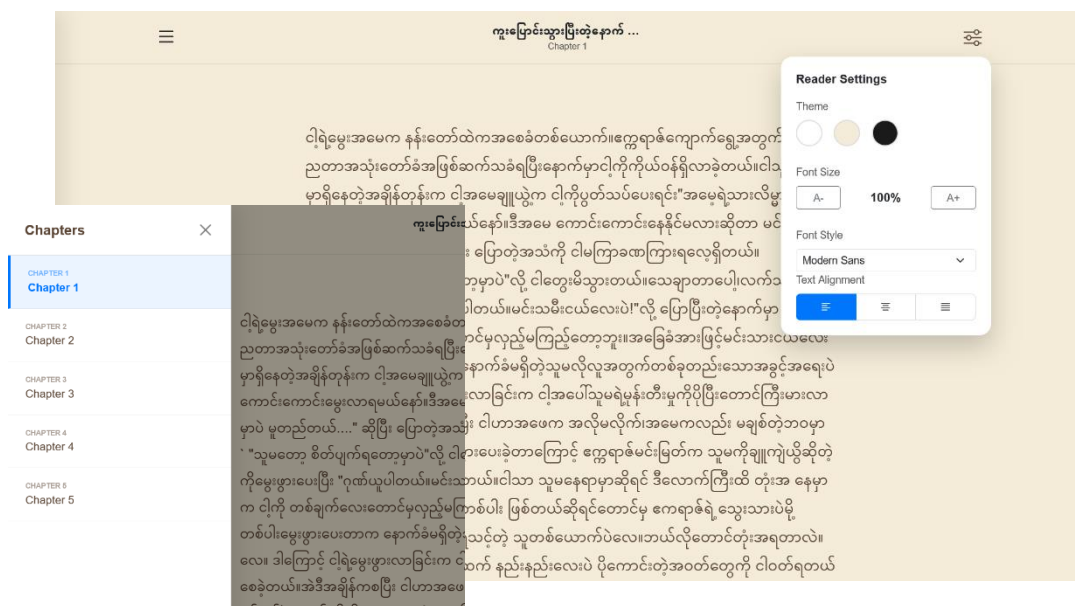


Figure 12: Read Chapters

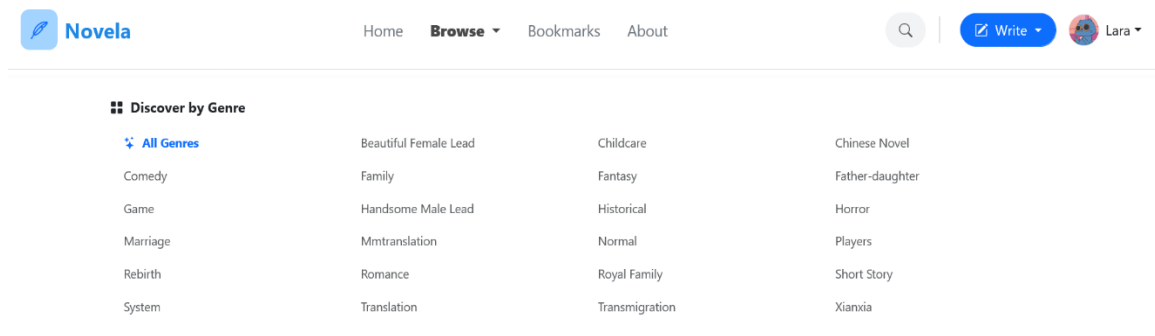


Figure 13: Browse Drop-down

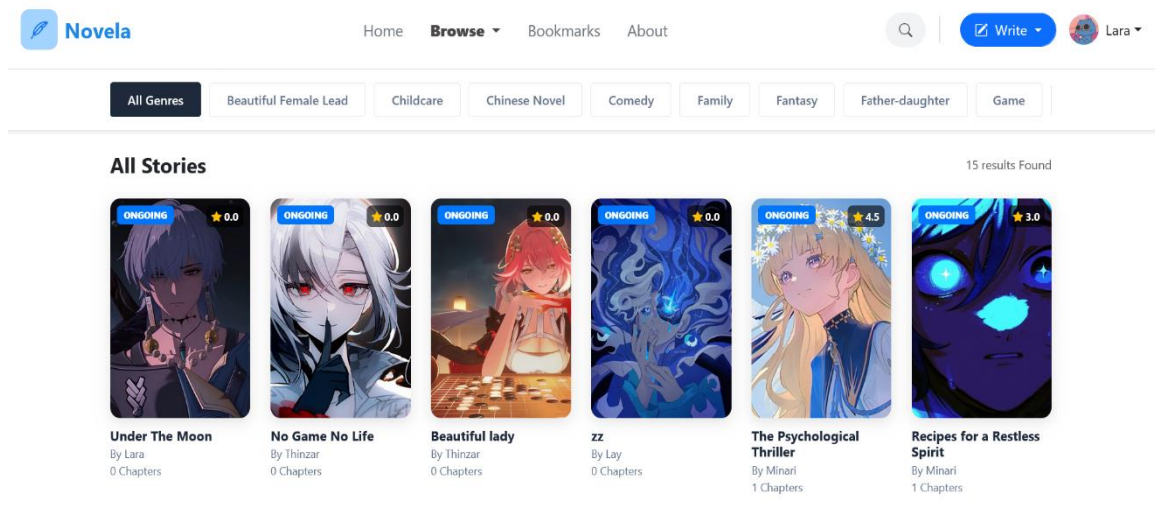


Figure 14: Browse Page

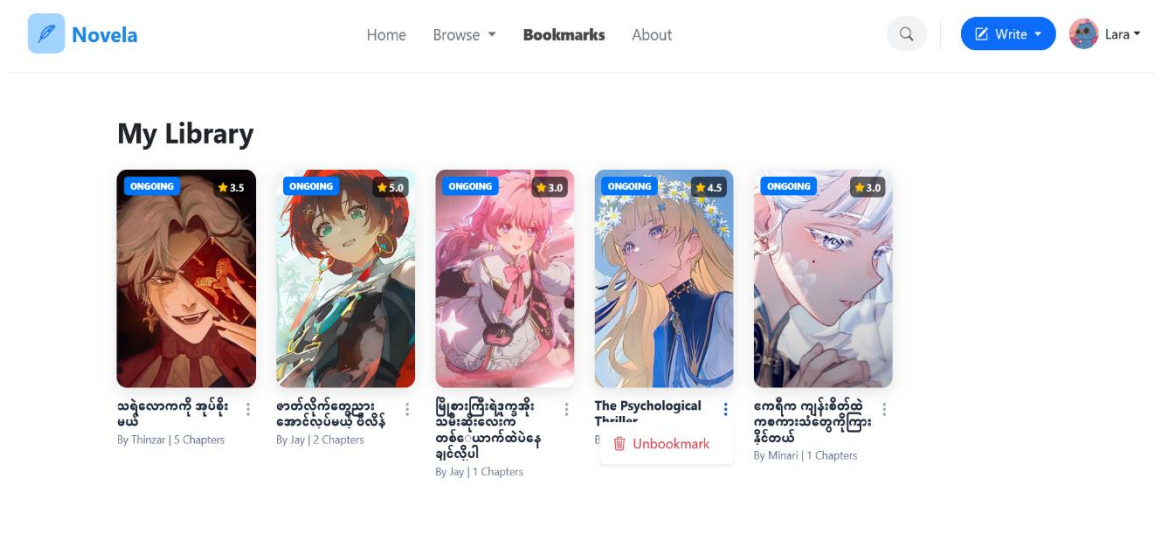


Figure 15: Bookmark Page

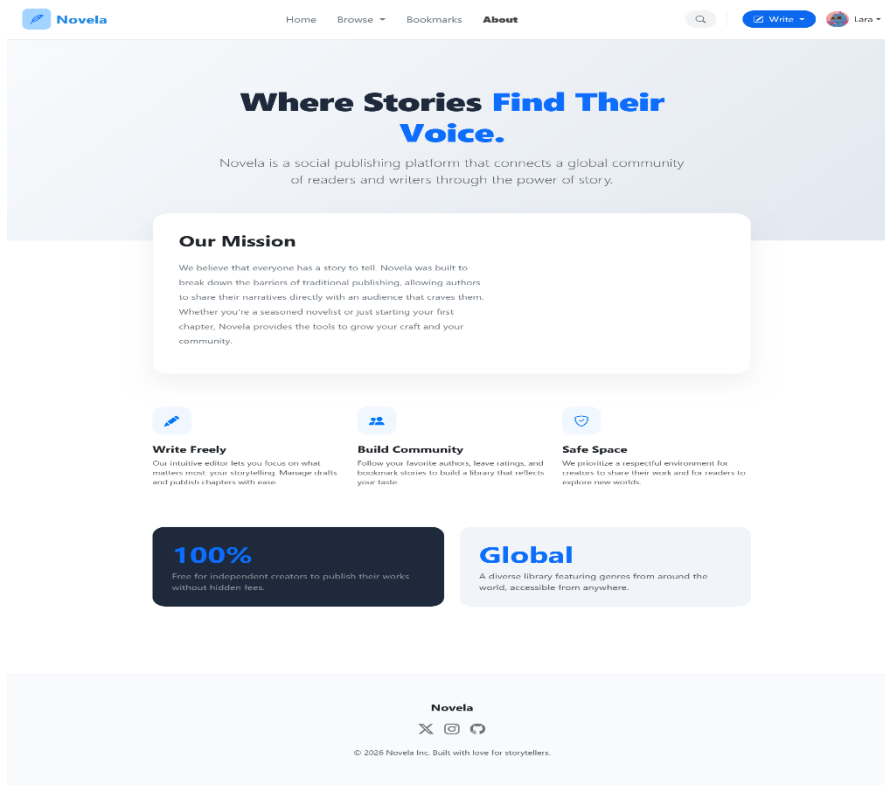


Figure 16: About Page

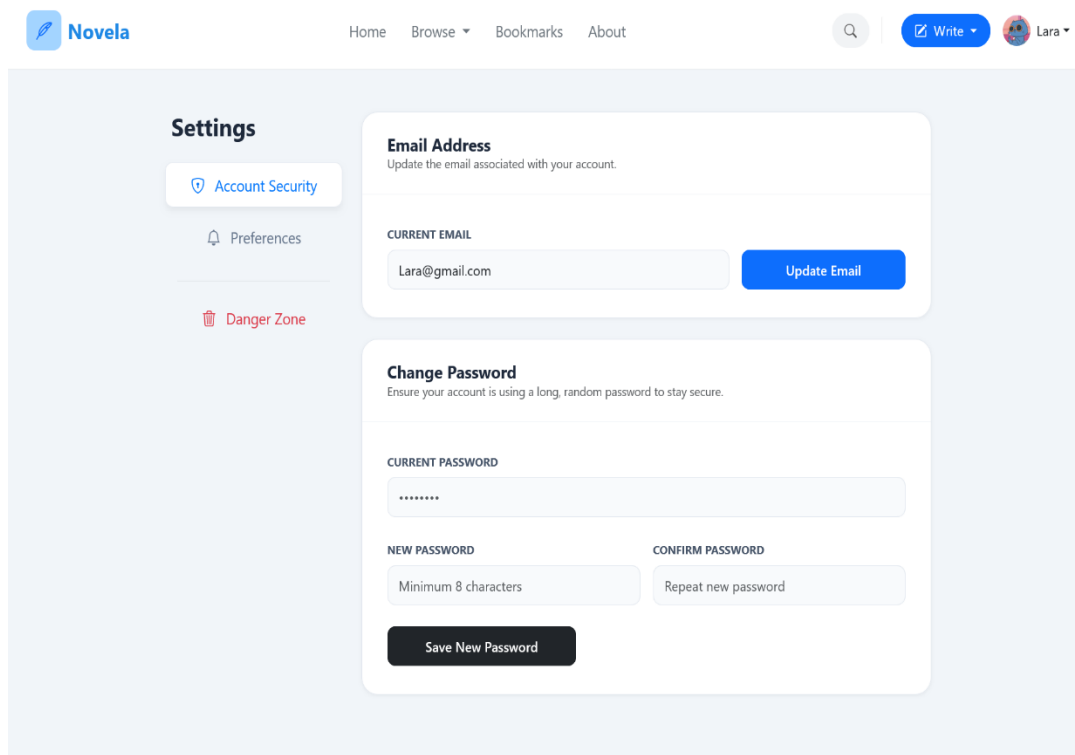


Figure 17: Settings Page


Novela

[Home](#)
[Browse](#)
[Bookmarks](#)
[About](#)

[Write](#)

L

Lay

[Admin Dashboard](#)
[My Profile](#)
[Settings](#)
[Logout](#)

My Library

Novela

Admin Dashboard

Dashboard

Manage Users

Reports 7

Exit Dashboard

TOTAL USERS

7

TOTAL NOVELS

17

PENDING REPORTS

7

User Management

User	Role Switcher	Status	Actions
Lay (Lay@gmail.com)	Master Admin	Active	
Bob (Bob@gmail.com)	User	Active	Suspend
Saw (saw@gmail.com)	User	Active	Suspend
Minari (Minari@gmail.com)	User	Active	Suspend
Lara (Lara@gmail.com)	User	Active	Suspend
Thinzar (Thinzar@gmail.com)	User	Active	Suspend
Jay (Jay@gmail.com)	User	Active	Suspend

Reported Content

Reporter	Novel Title	Reason	Actions
Bob	The Psychological Thriller	Inappropriate Content:	Delete Novel Dismiss
Jay	The Psychological Thriller	Inappropriate Content:	Delete Novel Dismiss
Jay	The Psychological Thriller	Harassment:	Delete Novel Dismiss
Jay	The Psychological Thriller	Plagiarism:	Delete Novel Dismiss
Jay	The Psychological Thriller	Harassment:	Delete Novel Dismiss
Jay	ကေရီက ကျန်းစိတ်ဆဲကကေားသံတွေကိုကြားနိုင်တယ်	Inappropriate Content:	Delete Novel Dismiss
Jay	ကေရီက ကျန်းစိတ်ဆဲကကေားသံတွေကိုကြားနိုင်တယ်	Spam:	Delete Novel Dismiss

Figure 18: Admin Page

Chapter-8

Conclusion

The Online Novel Reading and Writing System provides a structured, secure, and interactive digital platform for readers, writers, and administrators. The system integrates novel browsing, multi-chapter publishing, reading customization, bookmarking, user follow relationships, and rating functionality within a role-based access framework.

Through the use of requirement analysis, modeling techniques, and database-driven design, the project demonstrates the practical application of Software Engineering methodologies. Functional requirements such as authentication, content management, and administrative moderation are supported by non-functional requirements including performance optimization, security enforcement, reliability, usability, and maintainability.

The system successfully fulfills the defined business needs by delivering an organized and secure environment for digital literary content.

8.1 Future Work

Although the current system meets the defined requirements, future enhancements may include:

- Commenting system for novels
- Notification system for followers
- Advanced recommendation algorithms
- Mobile application version
- Advanced analytics dashboard for administrators
- Enhanced content reporting mechanisms

These improvements would increase platform engagement, scalability, and analytical capabilities while maintaining the existing architectural structure.

8.2 References

1. <https://www.wattpad.com/home>
2. <https://mmxianxia.com/>
3. https://www.measureevaluation.org/resources/publications/tl-19-34/at_download/document